

Decorating microbes: surface display of proteins on *Escherichia coli*

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Bacterial surface display entails the presentation of recombinant proteins or peptides on the surface of bacterial cells. *Escherichia coli* is the most frequently used bacterial host for surface display and, as such, a variety of *E. coli* display systems have been described that primarily promote the surface exposure of peptides and small proteins. By contrast, display systems based on autotransporter proteins (ATs) and ice nucleation protein (INP) are excellent systems for the display of large and complex proteins, and are therefore of considerable biotechnological relevance. Here, we review recent advances in AT and INP-mediated display and their biotechnological applications. Additionally, we discuss several promising alternative display methods, as well as novel bacterial host organisms.

Introduction

Various microorganisms are used as hosts for surface display of recombinant proteins and peptides, including Gram-positive bacteria and yeast [1]. However, the Gram-negative bacterium *Escherichia coli* is the most frequently used host primarily because of its capability to produce recombinant proteins in high yields, and amenability to genetic manipulation. To achieve presentation at the cell surface, a display system has to traverse the complex *E. coli* cell envelope, which is composed of both an inner membrane (IM) and an outer membrane (OM), which are separated by the periplasm. Surface display of the protein of interest (the passenger) is accomplished by genetically fusing it to a carrier protein, which facilitates export across the cell envelope and anchors the passenger to the bacterial cell surface. It is crucial that the carrier protein contains a robust surface anchor to attach the passenger to the cell surface and to ensure proper surface exposure. By being displayed on the cell surface, the passenger is free to access any externally added substrate, and membrane penetration of the targeted substrate is not an issue. This unrivalled accessibility has opened up avenues to several applications, ranging from protein library screening to the production of biofuels [2].

Although a variety of different carrier proteins have been described for the display of passengers on *E. coli* (Table 1), the use of these systems is primarily restricted to the surface exposure of peptides and small proteins. By

contrast, systems based on autotransporter proteins (ATs) and ice nucleation protein (INP) have proven to be outstanding carriers for the display of large and complex passengers (Tables 1 and 2), and are therefore of considerable biotechnological interest. This review deals mainly with recent advances in AT- and INP-mediated surface display and their applications in whole-cell biocatalysis and high-throughput screening of protein libraries (Box 1). In addition, we discuss several promising alternative display methods, as well as novel bacterial hosts (Box 2).

Overview of *E. coli* display systems

Systems based on outer membrane proteins (OMPs)

Different protein scaffolds have been used as carriers (Table 1). However, the vast majority is based on *E. coli* β -barrel OMPs, such as LamB, FhuA, and the porins OmpA, OmpC and OmpX [3–6]. β -Barrel proteins are integral membrane proteins, which span the OM with antiparallel β -sheets that form a barrel-like structure.

Most OMP-based display systems only tolerate the insertion of small peptides in surface-exposed loops, without a significant loss of stability (Figure 1a). To overcome these limitations, a novel display system based on the OmpX eCPX has been developed and optimized (Table 1). eCPX allows the attachment of passenger peptides to its N or C terminus, or both [5]. Several systems have been developed recently that promote the display of fairly large passengers. For example, a truncated variant of OmpC and a system based on the hypothetical OMP Omp1 from *Zymomonas mobilis* promote the display of C-terminally fused passengers of 50 kDa and 56 kDa, respectively [4,7]. Furthermore, the *Bacillus subtilis* protein PgsA localizes to the cell surface when expressed heterologously in *E. coli*. This has been exploited by using PgsA as a carrier, which enables the display of α -amylase (77 kDa) from *Streptococcus bovis* and lipase B (34 kDa) from *Candida antarctica* fused to its C terminus [8].

Systems based on surface appendages

The surface of several bacterial species is decorated with specialized protrusions, such as flagella, pili or fimbriae. A flagellum is made up of roughly 20 different proteins. The flagellar rod is primarily composed of the flagellin protein FliC, and the flagellar cap is formed by FliD. Both FliC and FliD have been used as carrier proteins that allow the display of small peptides inserted into exposed sites [6,9].

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Table 1. Typical surface display systems for *E. coli*

Scaffold	Type of fusion	Passenger size (kDa)	Application	Verification of surface display	Refs.
Outer membrane proteins					
eCPX derived from OmpX	Biterminal	0.8 – 1.6	Peptide library screening	Flow cytometry, fluorescence microscopy	[5]
FhuA	Insertional	1.1 – 3.3	Peptide library screening	Flow cytometry	[6]
LamB	Insertional	1.2 –25.5	Bioremediation, peptide library screening, vaccine development	Cell fractionation, immunoblotting, functional assays	[6]
Omp1	C-terminal	56	Biocatalysis	Cell fractionation, immunoblotting, functional assays	[7]
OmpA	Insertional	1 – 50	Peptide library screening, peptide display, vaccine development	Cell fractionation, immunoblotting, fluorescence microscopy, flow cytometry	[3,6,55,56]
OmpC	Insertional, C-terminal	18-52	Bioremediation, biocatalysis	Functional assays	[4,6]
OmpT		35	Directed evolution, substrate profiling	Flow cytometry	[57]
OprF	C-terminal	50	Biocatalysis	Cell fractionation, immunoblotting, fluorescence microscopy, functional assays	[58]
PgsA	C-terminal	34-77	Biocatalysis	Cell fractionation, immunoblotting, fluorescence microscopy, flow cytometry	[8]
Wza-omp orf1/OmpU/Omp26La	C-terminal	27-50	Translocation studies	Cell fractionation, immunoblotting, functional assays	[12]
Surface appendages					
F pillin	Insertional	1.6	Peptide antigen display	Functional assays	[6]
Fimbriae (FimH and FimA)	Insertional	1-4	Peptide library screening, immunogenic peptide display, bioremediation	Functional assays, fluorescence microscopy	[1]
Flagellin (FliC and FliD)	Insertional	1.2-33	Peptide display, peptide library screening, vaccines, bioremediation, exploring protein-protein interactions	Immunoelectron microscopy, flow cytometry	[6,9]
Lipoproteins					
INP	C-terminal	7-119	Epitope mapping, biocatalysis, vaccines, protein library screening	Cell fractionation, immunoblotting, flow cytometry, functional assays	[6,38,59]
Lpp-OmpA	C-terminal	27 – 74	Bioremediation, biocatalysis, antibody library screening	Cell fractionation, immunoblotting, fluorescence microscopy, functional assays	[6,10,11]
PAL	N-terminal	29	Display antibody fragments, bioremediation	Flow cytometry, functional assays	[6]
Tat-dependent lipoprotein	C-terminal	27	Translocation studies	Cell fractionation, immunoblotting, fluorescence microscopy	[20]
TraT	Insertional, C-terminal	1.2 – 11	Antigen display	Dot blot, fluorescence microscopy	[6]
Virulence factors					
AIDA-I	N-terminal	12-65	Biocatalysis, protein library screening, vaccine development	Cell fractionation, immunoblotting, fluorescence microscopy, flow cytometry, functional assays	[23,27,37]
EaeA	C-terminal	3.9 – 31.6	Translocation studies	Cell fractionation, immunoblotting, fluorescence microscopy, flow cytometry	[1,31]
EspP	N-terminal	20	Protein library screening	Cell fractionation, immunoblotting, flow cytometry	[23,37]
EstA	N-terminal	38-60	Biocatalysis, protein library screening	Flow cytometry, functional assays	[23,31,32,36]
Invasin	C-terminal	1.1	Peptide library screening	Flow cytometry, fluorescence microscopy	[1]
MSP1a	N-terminal	4.6	Immunogenic peptide display	Cellular fractionation, fluorescence microscopy, functional assays	[15]

The commercially available FliTrx system is based on flagellar display and was originally developed to study protein–protein interactions [6].

Fimbriae and pili are structurally similar to flagella and are composed of one or a few subunits. The major fimbrial subunits FimH and FimA, as well as the pillus subunit F pilin have been used as carriers. These subunits can be used for the display of heterologous peptides, which are inserted at exposed sites [1,6].

Lipoprotein-derived systems

Lipoproteins comprise another class of bacterial OMPs, which are anchored to the OM by virtue of their lipid-modified N terminus. Two lipoproteins have been used as carriers, namely TraT and PAL. TraT is a versatile carrier because it promotes the display of peptides inserted into the middle of the protein, as well as peptides attached to its C terminus [6]. By contrast, a system based on PAL enables the display of an N-terminally fused passenger [6]. Lpp

Table 2. Overview of passengers displayed by INP system

Source INP	Gene	Truncation ^a	INP size (kDa)	Passenger	Passenger size (kDa)	Cofactor	Refs.
<i>Pseudomonas syringae</i>	<i>inaK</i>	NC	33	AldO	45	FAD	[25]
<i>P. syringae</i>	<i>inaV</i>	NC	36	carboxylesterase	60		[60]
<i>P. syringae</i>	<i>inaK</i>	N	22	Chi92	90		[40]
<i>P. syringae</i>	<i>inaK</i>	wt & NC	114	CMCase	33		[6]
<i>P. syringae</i>	<i>inaK</i>	wt	114	CMCase library	33		[6]
<i>P. syringae</i>	<i>inaK</i>	NC	33	CPR	77	FAD and FMN	[39]
<i>P. syringae</i>	<i>inaQ</i>	N	18	GFP	27		[43]
<i>P. syringae</i>	<i>inaK</i>	NC	36	HepC virus core	7		[59]
<i>P. syringae</i>	<i>inaK</i>	wt & NC	114	HIV-1 gp120	60		[6]
<i>P. syringae</i>	<i>inaK</i>	wt	114	Levansucrase	47		[6]
<i>P. syringae</i>	<i>inaK</i>	NC	36	MerR	16		[61]
<i>P. syringae</i>	<i>inaV; inaK</i>	NC	36	OPH	49		[10,42,62]
<i>P. syringae</i>	<i>inaQ</i>	N	18	PBP	37		[43]
<i>P. syringae</i>	<i>inaK</i>	NC	33	P450 BM3	119	FAD, FMN and heme	[38]
<i>P. syringae</i>	<i>inaK</i>	NC	36	Salmobin	25		[6]
<i>Xanthomonas campestris</i>	<i>inaX</i>	NC	27	Transglucosidase	60		[41]

^aTruncation refers to INP: NC, only N- and C- terminal domain used; N, only N-terminal domain; wt, no truncation employed.

Abbreviations: AldO, alditol oxidase; Chi92, chitinase 92; CMCase, carboxymethylcellulase; CPR, rat NADPH-cytochrome P450 oxidoreductase; GFP, green fluorescent protein; HepC, hepatitis C virus; HIV-1 gp120, human immunodeficiency virus type 1 viral envelope glycoprotein; MerR, metalloregulatory protein; OPH, organophosphorus hydrolase; P450 BM3, cytochrome P450 BM3; PBP, phosphate binding protein.

(Braun's lipoprotein) is an abundant *E. coli* OM lipoprotein, and its signal sequence has been combined with a truncated OmpA variant to obtain a hybrid display system (Figure 1b). This system facilitates the surface presentation of larger passengers, such as β -lactamase (30 kDa), green fluorescent protein (GFP; 27 kDa), and a fusion of

methyl parathion hydrolase and GFP (60 kDa) [6,10,11]. Recently, a collection of novel display systems based on the signal sequence of the *Vibrio anguillarum* lipoprotein Wza and several *V. anguillarum* OMPs has been developed [12]. Four of these fusion proteins are able to facilitate the display of C-terminally fused GFP, as well as a viral capsid protein of 50 kDa.

Box 1. Surface display in combination with ultra-high-throughput FACS-based screening

In recent years, fluorescent-activated cell sorting (FACS) has become a powerful tool for the activity-based screening of mutant proteins from large libraries. FACS involves complex instruments that can analyze and sort small particles (cells, beads, emulsion droplets) at extremely high event rates (up to 10 000/s) according to their fluorescent properties.

Use of *E. coli* cell surface display and FACS technology is clearly an attractive option for screening large enzyme libraries for improved activity. Surface display dramatically increases the substrate accessibility of an enzyme because permeability barriers imposed by the cell envelope are no longer an issue. This accessibility is at the same time the potential drawback of this approach because the product is also freely diffusible, therefore, the challenge is to bind the product to the cell surface, thereby establishing the phenotype-genotype linkage that is essential in enzyme engineering.

A rather elegant approach that has been recently piloted involves the covalent attachment of the fluorescent product via tyramide signal amplification [35,47]. This has been used in a number of cases that involve the screening of peroxidase, lipase, protease and esterase libraries [34]. The power of this technique is that the (fluorescent) product becomes covalently linked to the cell surface. The fluorescently labeled tyrosinol reacts with horseradish peroxidase (HRP) (or some other peroxidase) and H₂O₂ to form a quinone-type radical. This radical then spontaneously couples to random tyrosine residues found in cell surface proteins in the near vicinity, covalently coupling a fluorescent signal to the cell surface, and enabling detection of cells that express active mutant enzymes [48]. Screening of lipases, proteases and esterases can be achieved by blocking the phenolic hydroxyl group of tyramide so that a cleavage site is introduced, for example, for an esterase. By decorating the cells with externally added HRP, only cells that express active esterase on their surface can convert the tyramide precursor into its active form, which results in its deposition on the cell surface. Drawbacks notwithstanding, we envision a powerful tool in the near future in the coupling of FACS with surface display.

Systems based on virulence factors

Several pathogenic bacteria express highly specialized virulence factors on their cell surface, which are often essential for the initiation and progress of a disease. Two bacterial virulence factors have been used successfully

Box 2. Alternative display strategies

The complex *E. coli* cell envelope and unstable OM pose challenging obstacles for the display passengers, which is reflected by the fact that, in many cases, display of foreign passengers on the *E. coli* cell surface hampers cell viability. Therefore, several alternative methods have been established, such as phage display and ribosome display, which circumvent the presentation of passengers on the surface of *E. coli*, and are described elsewhere [49,50]. A newly developed alternative strategy, anchored periplasmic expression, involves display of the passenger in the periplasm of *E. coli*, which is tethered to the periplasmic face of the IM via a lipoprotein-targeting motif [51]. This technique has proven to be a valuable tool in the engineering of antibodies [52]. For proteins that refrain from being translocated through the IM and/or OM, intracellular display methods have been established that rely on the attachment of the protein of interest to intracellularly co-produced polyhydroxyalkanoate granules or intracellular protein inclusion bodies. These target protein-loaded particles can be isolated via cell lysis and centrifugation, and further used for various biotechnological applications [53,54].

Other bacteria have been explored as alternative hosts for surface display, including *B. subtilis*, *Lactococcus lactis* and several non-pathogenic *Staphylococci* [1]. In addition, *Streptomyces coelicolor* appears a promising novel host as indicated by recent experimental evidence [19]. In *S. coelicolor*, many proteins are exported via the Tat pathway, including several cell-wall-associated lipoproteins. Therefore, *S. coelicolor* appears to be an ideal platform for the surface display of foreign (cofactor-containing) enzymes by means of the Tat pathway.

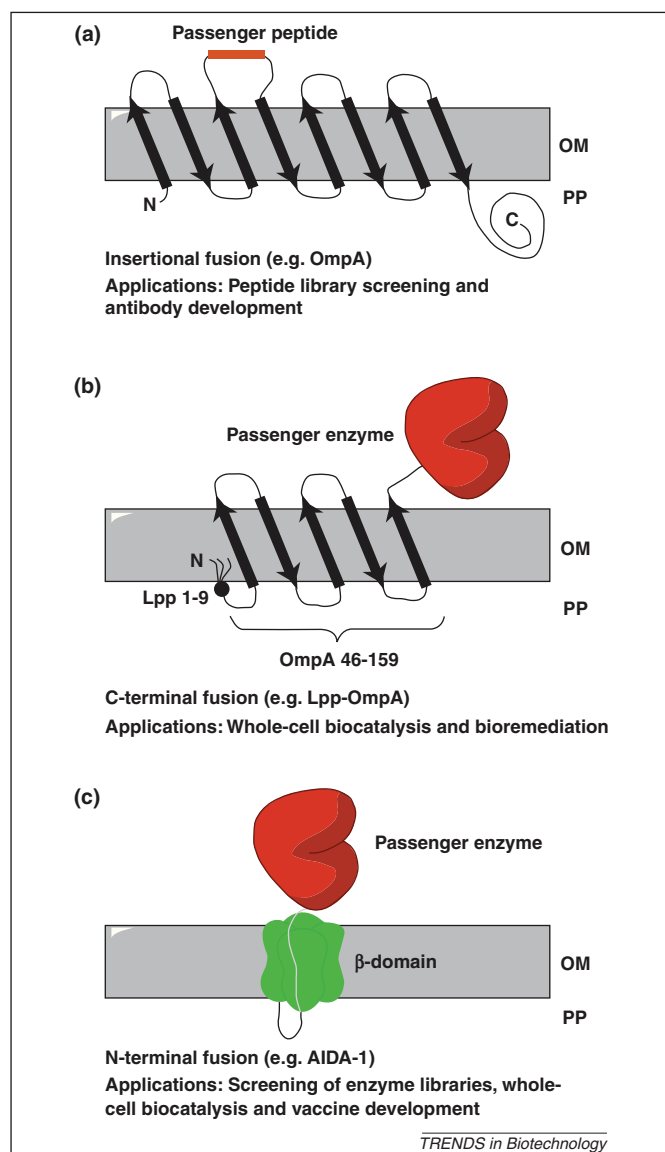


Figure 1. Examples of typical *E. coli* surface display systems. The passenger can be fused to a carrier at its N or C terminus, or as an insertional fusion. The passenger and carrier are separated by a spacer of appropriate length to avoid functional interference between the passenger and cell surface, or between the passenger and the carrier. **(a)** Schematic representation of OmpA as a prototype system for the display of a passenger peptide (orange) inserted into a surface exposed region. The transmembrane antiparallel β -sheets are shown as arrows. **(b)** Model of a hybrid display system that comprises the signal sequence of the lipoprotein Lpp, as well as the first nine residues of the mature protein and residues 46–159 of mature OmpA. This system promotes the display of a passenger enzyme (orange) attached to its C terminus. The Lpp and OmpA portions are indicated and OmpA transmembrane antiparallel β -sheets are shown as arrows. **(c)** Cartoon structure of an AT-based display scaffold used for the surface presentation of an N-terminally fused passenger enzyme (orange). Abbreviations: PP, periplasm.

as carrier proteins, namely invasins and the intimin protein EaeA. Invasin is a protein of 106 kDa that resides in the OM of *Yersinia pseudotuberculosis*. The N-terminal portion of invasins anchors the protein to the OM and forms a β -barrel, whereas the surface-exposed C-terminal domain is a rod-like structure and is made up of five tandem sub-domains [13]. It has been shown that an invasins derivative that comprises the OM domain can be employed as carrier for the display of peptides fused to its C terminus [6]. The intimin protein EaeA is an OMP of 102 kDa and is found in enteropathogenic *E. coli*. The N-terminal domain of EaeA

is required for OM anchoring, and the extracellular C-terminal stalk is composed of four tandem sub-domains, as well as a receptor sub-domain [13]. Moreover, it has been demonstrated that a truncated EaeA variant, which lacks the receptor and two tandem domains, enables the display of C-terminally fused passenger proteins, ranging from human interleukin-4 (17 kDa) to β -lactamase (30 kDa) [1,14]. Recently, a carrier derived from the major surface protein 1a from *Anaplasma marginale* has been described [15]. This system enables the surface presentation of immunogenic peptides at the surface of *E. coli* by means of an N-terminal fusion. Taken together, these studies show that these bacterial virulence factors are good alternatives to β -barrel OMP-based systems for the display of peptides and small proteins, owing to their unique and modular structure.

Tat-dependent systems

The vast majority of periplasmic and OMPs in *E. coli* are exported out of the cytoplasm in an unfolded state via the IM-embedded Sec-translocon, which functions as a protein-conducting channel [16]. Most known display systems also rely on the Sec-translocon for periplasmic export; however, by contrast, the Tat system is dedicated to the periplasmic export of fully folded and often cofactor-containing substrate proteins [17]. Evidence is accumulating that the Tat system is required for the biogenesis of several bacterial OMPs, including lipoproteins [18–20]. This finding has recently been used to develop a novel *E. coli* display system, which combines key elements of a prototype Tat-dependent signal sequence and a lipoprotein-type signal sequence [20]. Furthermore, PnlH represents a Tat-dependent OMP of *Dickeya dadantii*. PnlH contains an uncleaved signal sequence, and it has been shown that this signal sequence is able to target foreign proteins to the *E. coli* OM [18]. However, these hybrid proteins are not presented at the cell surface of *E. coli*; probably because the specialized machinery for OM translocation is not present. These data show that the PnlH signal sequence contains all the information required to export passengers to the OM Tat-dependently, and therefore, might serve as a basis for the design of a novel display system.

Autotransporter proteins

The term ‘autotransporter’ was coined for a family of proteins that carry sufficient information on a single gene to direct their own secretion out of a bacterial cell. In most cases, ATs retain their ability to insert into the OM and to translocate a passenger domain through the OM when transferred to other bacterial hosts. Most ATs are virulence factors of pathogenic Gram-negative bacteria. These proteins consist of a single polypeptide that is composed of two domains: a passenger domain that is transported across the cell envelope, and a translocator domain that mediates the OM translocation step. ATs contain an N-terminal signal sequence, and translocation across the IM occurs via the Sec pathway [21]. Despite their importance, the mechanism by which passenger domains of ATs are translocated across the OM is still under debate [21]. ATs form an efficient system for the display of foreign enzymes because the native passenger domain can easily be

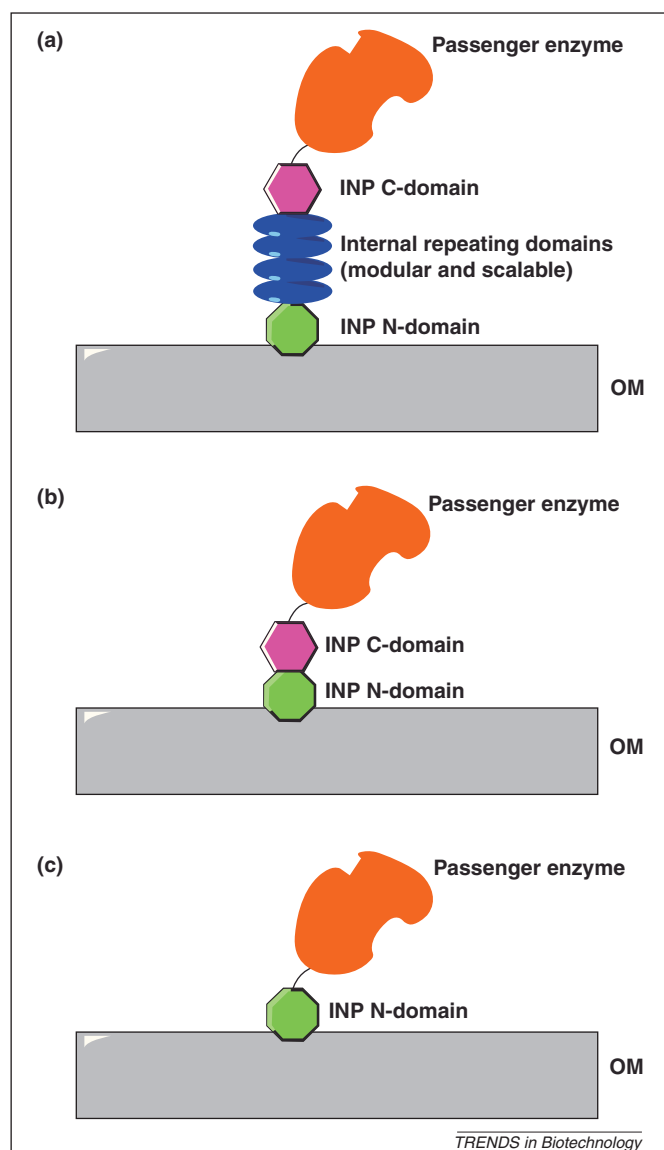


Figure 2. Schematic representation of cell surface display using INP. Although the biochemistry, genetics and applications of bacterial INPs have been studied in some detail, no signal sequence has been identified, and the precise mode of export remains unknown. Depicted are the various truncations possible. (a) INP with no truncation or truncation of some of the internal repeating domains. (b) INP N and C domains with truncation of the entire internal repeating domain. (c) INP N domain and surface display with only the N-terminal anchoring domain. Passenger enzymes are shown in orange.

replaced by the protein of interest. As such, these systems promote the display of N-terminally fused passengers (Figure 1c). Another aspect that contributes to the popularity of these display systems is their relatively high cell-surface expression on *E. coli*, which can reach a level comparable to that of several major endogenous porins, such as OmpA with about 100 000 copies per cell [22,23].

Ice nucleation protein

INP is an OMP that is found in several plant pathogenic bacteria [24]. INP has several unique structural and functional features that make it highly suitable for use in a bacterial surface display system. The specific amino acids of the N-terminal domain are relatively hydrophobic and link the protein to the OM via a glycosylphosphatidylinositol anchor [24]. The C-terminal domain of the protein is

highly hydrophilic and exposed to the medium. The central part of INP comprises a series of repeating domains that act as templates for ice crystal formation [24].

It has been shown that full-length INP and various truncates that lack the central repeating domain yield stable surface display (Table 2). This indicates that the central repeating domains are not required for export to the cell surface, and are therefore, ideal spacer units to control the distance between the passenger protein and the cell surface (Figure 2a). A derivative that comprises the N- and C-terminal domains is commonly used for surface display (Figure 2b). However, the N-terminal domain appears to be the only prerequisite for successful targeting and surface-anchoring (Figure 2c and Table 2). Importantly, INP can be expressed at the cell surface of *E. coli* at a very high level, without affecting cell viability: comparable to the endogenous expression of the OmpA porin [22,25].

Autodisplay and applications

The use of AT-based systems for the surface presentation of foreign enzymes was first described in 1997 [23]. Since then, this autodisplay technology has been used in various biotechnological applications, including the construction of different whole-cell biocatalytic systems. The first AT to be described was IgA protease of *Neisseria gonorrhoeae* [26]. The IgA protease C-terminal domain (Iga β) has been used successfully for the display of different foreign proteins, such as cysteine-rich small protease inhibitors, a leucine zipper, and a single chain antibody domain [23]. Importantly, it has been shown that the presence of disulfide bonds within some passenger proteins impairs Iga β -mediated display. This could be prevented by using an *E. coli* *dsbA* null strain, which indicates that OM translocation of the passenger requires an unfolded and disulfide-free conformation [23].

An *E. coli* adhesin, AIDA-I, has been widely used for the display of various passenger domains [23]. The natural protein is synthesized as a 78-kDa adhesin passenger domain and a 45-kDa translocator domain. This translocator domain has been used for the display of large passengers, such as a fusion between organophosphorus hydrolase (OPH) and GFP (82 kDa) [27], sorbitol dehydrogenase (29 kDa) [28], β -lactamase (30 kDa) [29] and the esterase ApeE (67 kDa) [30]. Likewise, the *Pseudomonas aeruginosa* AT EstA has been used successfully for the display of enzymes, such as members of the lipase and esterase family. Flow cytometry analysis and measurement of lipase activity has revealed that *B. subtilis* lipase LipA (19 kDa), *Fusarium solani* pisi cutinase (23 kDa), and *Serratia marcescens* lipase (67 kDa), which is one of the largest lipases currently known, are all efficiently displayed by EstA, while retaining their lipolytic activities [31]. Moreover, the closely related EstA AT from *Pseudomonas putida* has been shown to promote the display of the industrially important *Pseudomonas* and *Burkholderia* lipases. Both of these enzymes require a lipase-specific foldase for their proper folding, which can be co-expressed in *E. coli* [32]. The resulting whole-cell systems are interesting biocatalytic tools for enantioselective conversions. Moreover, the recent use of AT-based display systems in

biosensors and immunoassays emphasizes their versatility [30,33].

Autodisplay and high-throughput library screening

In several experimental approaches, it has been shown that EstA display provides a useful tool for cell surface display and ultra-high-throughput screening of variant libraries generated by directed evolution, thereby enabling the identification of novel enzymes with interesting biological and biotechnological applications [34–36].

Very recently, the display efficiencies of small, engineered, single-chain binding proteins have been compared based on the lipocalin scaffold (Anticalins) with respect to display level, functional variance and bacterial cell viability [37]. The β -domains of five different bacterial ATs have been considered: IgA protease; EstA; the YpjA autotransporter from *E. coli* K12; the AIDA-I adhesin; and the protease EspP from enterohemorrhagic *E. coli*. The EspP autotransporter displays high genetic stability and mediates the presentation of fully functional passengers at high density on the *E. coli* cell surface. Affinity maturation of a cytotoxic-T-lymphocyte-antigen-4-binding Anticalin was successfully performed via surface display of a combinatorial Anticalin library, and repeated cycles of incubation with the fluorescently labeled target protein and fluorescence-activated cell sorting. It has yet to be proven whether EspP display is superior to other AT systems when other passenger proteins other than Anticalins are used.

INP-mediated display and applications

INP is frequently used to develop whole-cell biocatalysts (Table 2), primarily because: (i) INP does not appear to be hampered by the size of the passenger; and (ii) INP is compatible with the translocation and surface display of proteins that contain multiple cofactors as well as disulfide-bond-containing passengers.

Cofactor-containing enzymes

These important features have been illustrated well by a recent study [38], in which the 119-kDa cytochrome P450 BM3 from *Bacillus megaterium* was fused to the N- and C-terminal domains of INP, and whole-cell biocatalysis was achieved with similar substrate specificity and rates to that of purified enzyme. The impressive size of the protein successfully expressed on the cell surface and the fact that it contains several cofactors are noteworthy. Two other studies have revealed the power of the INP system in transporting cofactor-containing enzymes. In the first study, the diflavin-containing rat NADPH-cytochrome was displayed on the surface of *E. coli* [39]. Similarly, in a separate study, *Streptomyces coelicolor* alditol oxidase, a covalent flavoprotein oxidase, was displayed [25]. In all cases, the cofactors are incorporated in the cytoplasm, which indicates that the mode of export of INP is compatible with fully folded, cofactor-containing proteins.

Other enzymes

INP-derived scaffolds are commonly used for the surface exposure of other enzymes, which yields a variety of whole-cell biocatalysts (Table 2). To obtain a whole-cell system

that can be used for biological control against phytopathogenic fungi, chitinase 92 from *Aeromonas hydrophila* JP10 was displayed on the surface of *E. coli* by means of a truncated INP derivative [40]. The resultant whole-cell system was able to inhibit the mycelial growth of various phytopathogenic fungi. Furthermore, a whole-cell biocatalyst that can be utilized in glucosylation reactions has been constructed via INP-mediated surface display of *Xanthomonas campestris* transglucosidase [41].

INP-mediated display has been used to develop several potential tools in bioremediation. Recently, an INP-based display system has been used for the presentation of OPH on the surface of *E. coli*, while an Lpp–OmpA system has been used for the display of enhanced GFP on the surface of the same cells [10]. This co-display has resulted in an autofluorescent whole-cell system that is able to degrade organophosphates, potentially *in situ*. To engineer an *E. coli* strain with a broader substrate scope than a strain that expresses OPH alone, OPH and methyl parathion hydrolase (MPH) have been co-expressed in the same host; either at the cell surface or the periplasm [42]. INP-mediated display has been used to present OPH at the cell surface, whereas MPH has been directed to the periplasm by fusing it to a Tat-dependent signal sequence. The resultant recombinant strain is highly active against a variety of organophosphorous pesticides. Moreover, this system appears remarkably robust because almost all of the whole-cell activity is retained over a period of 2 weeks. Therefore, this whole-cell system is a promising tool for the detoxification of organophosphates.

Likewise, INP-mediated surface display of phosphate-binding protein onto the surface of *E. coli* or *P. putida* has resulted in whole-cell systems that can be used for the removal of eutrophic phosphorus from wastewater, as judged by their ability to scavenge dibasic and monobasic phosphates over time. Remarkably, upon heat inactivation, these cells retain to a large extent their phosphate-scavenging capacity [43]. Taken together, these studies emphasize the versatility and robustness of INP-based display systems. Additionally, INP appears to be an excellent scaffold for the display of passengers that contain multiple disulfide bonds, such as the thrombin-like enzyme salmabin, which contains six potential disulfide bonds, and the major HIV envelope glycoprotein gp120, which contains nine potential disulfide bonds (Table 2). Hence, the tolerance and flexibility of INP-type scaffolds ensures their use as a popular display system in diverse biocatalytic applications.

Conclusions and perspective

Although bacterial surface display has made important contributions to the design of whole-cell biocatalysts and high-throughput library screening (Box 1), several challenges remain. These include the display of cofactor-containing enzymes, the co-display of more than one enzyme, and the display of large functional protein complexes.

The display of cofactor-containing enzymes remains cumbersome, primarily because established display systems are of limited use for these enzymes. However, several alternative display systems have been described, which are well-suited for this task. These include INP-

type scaffolds and a recently described system that combines key elements of a Tat-dependent signal sequence and a lipoprotein-type signal sequence [20]. However, it needs to be established whether this Tat-dependent display system can be utilized for the surface presentation of cofactor-containing enzymes. By contrast, INP-type display systems are excellent scaffolds for the presentation of these enzymes [25,38,39]. The simultaneous use of multiple display systems appears to be a good strategy for the co-display of more than one protein [10]. However, it is questionable whether this strategy can be extended to the display of large protein complexes without detrimental secretion stress owing to the competition or overload of export pathways. This raises the question of which alternative approach is suitable for the display of large protein complexes. With respect to this issue, the fusion of several enzymes followed by INP-mediated surface display seems to be a promising alternative strategy because INP is able to facilitate the display of large passenger proteins of up to 119 kDa [38]. The increasing use of bacterial display technology will fuel the design of versatile and robust display systems, and it is therefore expected that these new systems will provide novel tools for the co-display of enzymes and large protein complexes.

Several successful strategies have been presented to increase the export of foreign proteins on *E. coli*, including the overexpression of chaperones, translocon components, and the optimization of signal sequences [44]. Conceivably, the growing understanding of protein translocation pathways and assembly of β -barrel OMPs, combined with available structural data, will provide novel leads for the redesign and optimization of protein translocation and assembly machineries, which in turn will be essential to increase and fine-tune the amount of the surface-displayed passenger. Specifically, future studies could be aimed at optimization of the signal sequence of a display system as well as the translocon (Sec or Tat) that it utilizes, to increase the periplasmic translocation of the passenger. For β -barrel OMPs and AT-based systems, it might be beneficial to optimize or overexpress the Bam complex – located in the OM and required for the OM assembly of β -barrel proteins [45] – to increase the amount of surface-displayed passenger.

Additionally, the combination of bacterial display and the ease with which *E. coli* cells can be genetically manipulated is particularly powerful, because it expands the molecular toolbox available to redesign and exploit metabolic pathways and to confer new enzymatic activities, as recently demonstrated for the production of biodiesel, fatty alcohols and waxes from simple sugars [46]. Immobilized enzymes are commonly employed to catalyze multi-step transformations (cascade reactions), which are often difficult to achieve by traditional chemical means. In this respect, it is particularly interesting to note that bacterial surface display is a potential alternative over the use of immobilized enzymes, because it allows the design of, for example, effective cascade reactions by employing bacterial cells that display a combination of the appropriate enzymes that can catalyze the different steps of the intended multi-step conversion.

Bacterial surface display represents an indispensable tool in biotechnology and it is expected that the growing use of surface display in combination with a detailed understanding of protein translocation pathways will lead to the design of novel systems as illustrated by the bi-terminal display scaffold eCPX and a system based on a Tat-dependent lipoprotein [5,20]. These new systems will be valuable tools and will help to solve bottlenecks in co-display and display of large protein complexes, which will result in robust display technology for industrial applications. Furthermore, the increasing use of surface display in the screening of massive enzyme libraries (Box 1) will result in highly sought after novel screening procedures, thereby opening up new avenues in the directed evolution of industrially relevant enzymes.

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